DBI2526 - Summer Project

1. Idea & Research Questions

1.1. Idea

Analysis of player data from the MMORPG game Final Fantasy XIV: A Realm Reborn.

1.2. Research Questions

1) How are players distributed between grand companies on a specific realm? (slice)
2) How many players are located on selected realms and are in the endgame? (dice)
3) How many realms are located on European servers? (drill)

2. Data

2.1. Sources

2.1.1. FFXIVCensus

Data from https://ffxivcensus.com was used for the following raw .csv:

- American_Realms.csv → https://ffxivcensus.com/#rat-na
- Japanese_Realms.csv → https://ffxivcensus.com/#rat-jp
- American_Realms.csv → https://ffxivcensus.com/#rat-eu
- Japanese_Realms.csv → https://ffxivcensus.com/#rat-oc
- American_Realms_Endgame.csv → https://ffxivcensus.com/#ra-na
- Japanese_Realms_Endgame.csv → https://ffxivcensus.com/#ra-jp
- American_Realms_Endgame.csv → https://ffxivcensus.com/#ra-eu
- Japanese_Realms_Endgame.csv → https://ffxivcensus.com/#ra-oc
- GrandCompanies.csv → https://ffxivcensus.com/#grandcompany (All Characters)
- Species_Sex.csv → https://ffxivcensus.com/#racegender (All Characters)
- Species_Sex_Endgame.csv → https://ffxivcensus.com/#racegender (Endgame Characters)
- Tribal_Interactions.csv → https://ffxivcensus.com/#tribal

Furthermore the values for Total Population and Endgame Characters was taken from: https://ffxivcensus.com/#population

2.2. Transformations

2.2.1. Dimensions

Dim_isEndgame

- 'Dim_isEngame' is a boolean value

Dim_GrandCompany

- 'Dim_GrandCompany' consists of 3 Grand Companies and the lack of one.

Dim_Realms

- 'Dim_Realms' concatenates all the 'Category' of American_Realms, European_Realms, Japanese_Realms and Oceanian_Realms.
- 'Category' is renamed to 'Dim_Realms'

Sub_Regions

- 'Dim_Realms' concatenates all the 'Category' of American_Realms, European_Realms, Japanese_Realms and Oceanian_Realms.
- 'Category' is renamed to 'Dim_Realms'
- Adds 'V_Region' which contains their Server Locations / Intended Regions.

2.2.2. Enrichment Dimensions

Dim_Species

- Copies 'Category' from Species_Sex_Engame.csv. The Endgame version is used since it is cleaner.
- 'Category' is renamed to 'Dim_Species'.

Sub_Sex

- 'Sub_Sex' consists of 2 sexes: male and female.

Dim_Tribes

- Copies 'Category' from Tribal_Interactions.csv.
- 'Category' is renamed to 'Dim_Tribes'.
- 'Tribe' is renamed to 'V_Interactions'.

Sub_Tribe_Categories

- 'Dim_Tribes' is copied from Dim_Tribes.
- Every Dim_Tribe gets a furry subcategory assigned, which was evaluated by professional furries. Said categories are in the column 'V_Category'.

2.2.3. Fact Tables

Fact

The base fact table shows the player amount per realm, grand company, and endgame status.

Steps taken to achieve this table:

- Realm-population numbers for total and endgame players as a calculation base
- Multiply by the percentile distribution of grand companies, which is calculated with the grand company population amount and the total population amount.

Enriched Fact

The base fact table shows the player amount per realm, grand company, endgame status, and species.

Steps taken to achieve this table:

- Realm-population numbers for total and endgame players as a calculation base
- Multiply by the percentile distribution of grand companies, which is calculated with the grand company population amount and the total population amount.
- Multiply by the percentile distribution of species, which was calculated with the total species population and the total population, or in the case of endgame characters, with the endgame species population and the endgame population.

Ridiculous Fact

The base fact table shows the player amount per realm, grand company, endgame status, and species. Furthermore, it also shows how many people have completed the Tribe-Interactions-Quests for the same attributes.

Steps taken to achieve this table:

- Realm-population numbers for total and endgame players as a calculation base
- Multiply by the percentile distribution of grand companies, which is calculated with the grand company population amount and the total population amount.
- Multiply by the percentile distribution of species, which was calculated with the total species population and the total population or, in the case of endgame characters, with the endgame species population and the endgame population.
- Multiply by the percentile distribution of completed quests, which was calculated with the total players who completed said quests per tribe and the total population.

3. Enrichment Idea

3.1. Ridiculous Research Questions

1) Which furry subcategory is most popular according to player data regarding the Tribes leveling mechanic?

4. Cube Visualization

4.1. Facts

4.1.1. Fact

Fact.csv

data > fact > Fact.csv

	Dim_isEndgame	Dim_Realms	Dim_GrandCompany	V_PlayerCount
1	True	Adamantoise	Immortal Flames	2440
2	True	Adamantoise	Maelstrom	2505
3	True	Adamantoise	Order of the Twin Adder	2955
4	True	Adamantoise	none	18147
5	True	Balmung	Immortal Flames	2356
6	True	Balmung	Maelstrom	2419
7	True	Balmung	Order of the Twin Adder	2854
8	True	Balmung	none	17525
9	True	Behemoth	Immortal Flames	2381
10	True	Behemoth	Maelstrom	2444

4.1.2. Enriched Fact

Enriched_Fact.csv

data > fact > Enriched_Fact.csv

	Dim_isEndgame	Dim_Realms	Dim_GrandCompany	Dim_Species	V_PlayerCount
1	True	Adamantoise	Immortal Flames	Au Ra	22
2	True	Adamantoise	Immortal Flames	Elezen	5
3	True	Adamantoise	Immortal Flames	Hrothgar	3
4	True	Adamantoise	Immortal Flames	Hyur	19
5	True	Adamantoise	Immortal Flames	Lalafell	16
6	True	Adamantoise	Immortal Flames	Miqo'te	32
7	True	Adamantoise	Immortal Flames	Roegadyn	4
8	True	Adamantoise	Immortal Flames	Viera	17
9	True	Adamantoise	Maelstrom	Au Ra	23
10	True	Adamantoise	Maelstrom	Elezen	5

4.1.3. Ridiculous Fact

Ridicilous_Fact.csv

data > fact > Ridicilous_Fact.csv

	Dim_isEndgame	Dim_Realms	Dim_GrandCompany	Dim_Species	Dim_Tribe	V_PlayerCount	V_TribeCompletions
1	True	Adamantoise	Immortal Flames	Au Ra	Kobold	22	1
2	True	Adamantoise	Immortal Flames	Au Ra	Sahagin	22	0
3	True	Adamantoise	Immortal Flames	Au Ra	Amaljaa	22	0
4	True	Adamantoise	Immortal Flames	Au Ra	Sylph	22	1
5	True	Adamantoise	Immortal Flames	Au Ra	Ixal	22	0
6	True	Adamantoise	Immortal Flames	Au Ra	Vanu Vanu	22	0
7	True	Adamantoise	Immortal Flames	Au Ra	Vath	22	0
8	True	Adamantoise	Immortal Flames	Au Ra	Moogle	22	0
9	True	Adamantoise	Immortal Flames	Au Ra	Kojin	22	0
10	True	Adamantoise	Immortal Flames	Au Ra	Ananta	22	1
11	True	Adamantoise	Immortal Flames	Au Ra	Namazu	22	0
12	True	Adamantoise	Immortal Flames	Au Ra	Pixie	22	1
13	True	Adamantoise	Immortal Flames	Au Ra	Qitari	22	0
14	True	Adamantoise	Immortal Flames	Au Ra	Dwarf	22	0

4.2. Dimensions

4.2.1. Dim_isEndgame

Dim_isEndgame.csv

data > transformed > Dim_isEndgame.csv

	Dim_isEndgame
1	false
2	true

4.2.2. Dim_Realms

Dim_Realms.csv

data > transformed > Dim_Realms.csv

1	Dim_Realms
2	Adamantoise
3	Balmung
4	Behemoth
5	Brynhildr
6	Cactuar
7	Coeurl
8	Diabolos
9	Excalibur
10	Exodus

4.2.3. Dim_GrandCompany

Dim_GrandCompany.csv

data > transformed > Dim_GrandCompany.csv

1	Dim_GrandCompany
2	Maelstrom
3	Immortal Flames
4	Order of the Twin Adder
5	None

4.2.4. Dim_Species

Dim_Species.csv

data > transformed > Dim_Species.csv

1	Dim_Species
2	Au Ra
3	Elezen
4	Hrothgar
5	Hyur
6	Lalafell
7	Miqo'te
8	Roegadyn
9	Viera

4.2.5. Dim_Tribes

Dim_Tribes.csv

data > transformed > Dim_Tribes.csv

	Dim_Tribe	V_Interactions
1	Kobold	842415
2	Sahagin	567836
3	Amaljaa	341675
4	Sylph	760943
5	Ixal	316697
6	Vanu Vanu	517775
7	Vath	535590
8	Moogle	507630
9	Kojin	621437
10	Ananta	673686

4.3. Subdimensions

4.3.1. Sub_Regions

Sub_Regions.csv

data > transformed > Sub_Regions.csv

1	Dim_Realms	V_Region
2	Adamantoise	Americas
3	Balmung	Americas
4	Behemoth	Americas
5	Brynhildr	Americas
6	Cactuar	Americas
7	Coeurl	Americas
8	Diabolos	Americas
9	Excalibur	Americas
10	Exodus	Americas

4.3.2. Sub_Sex

Sub_Sex.csv

data > transformed > Sub_Sex.csv

1	Dim_Sex
2	Male
3	Female

4.3.3. Sub_Tribe_Categories

Sub_Tribe_Categories.csv

data > transformed > Sub_Tribe_Categories.csv

1	Dim_Tribe	V_Category
2	Kobold	Furry
3	Sahagin	Auquarian
4	Amaljaa	Scaley
5	Sylph	Fey
6	Ixal	Avian
7	Vanu Vanu	Avian
8	Vath	Scaley
9	Moogle	Fey
10	Kojin	Scaley

5. Ridiculous Dimensions

See Enrichment Dimensions in Data

6. GeoData

GeoData is plotted according to total players per region.

7. Report on Research Questions

7.1. Research Questions

7.1.1. Question 1

Discussion

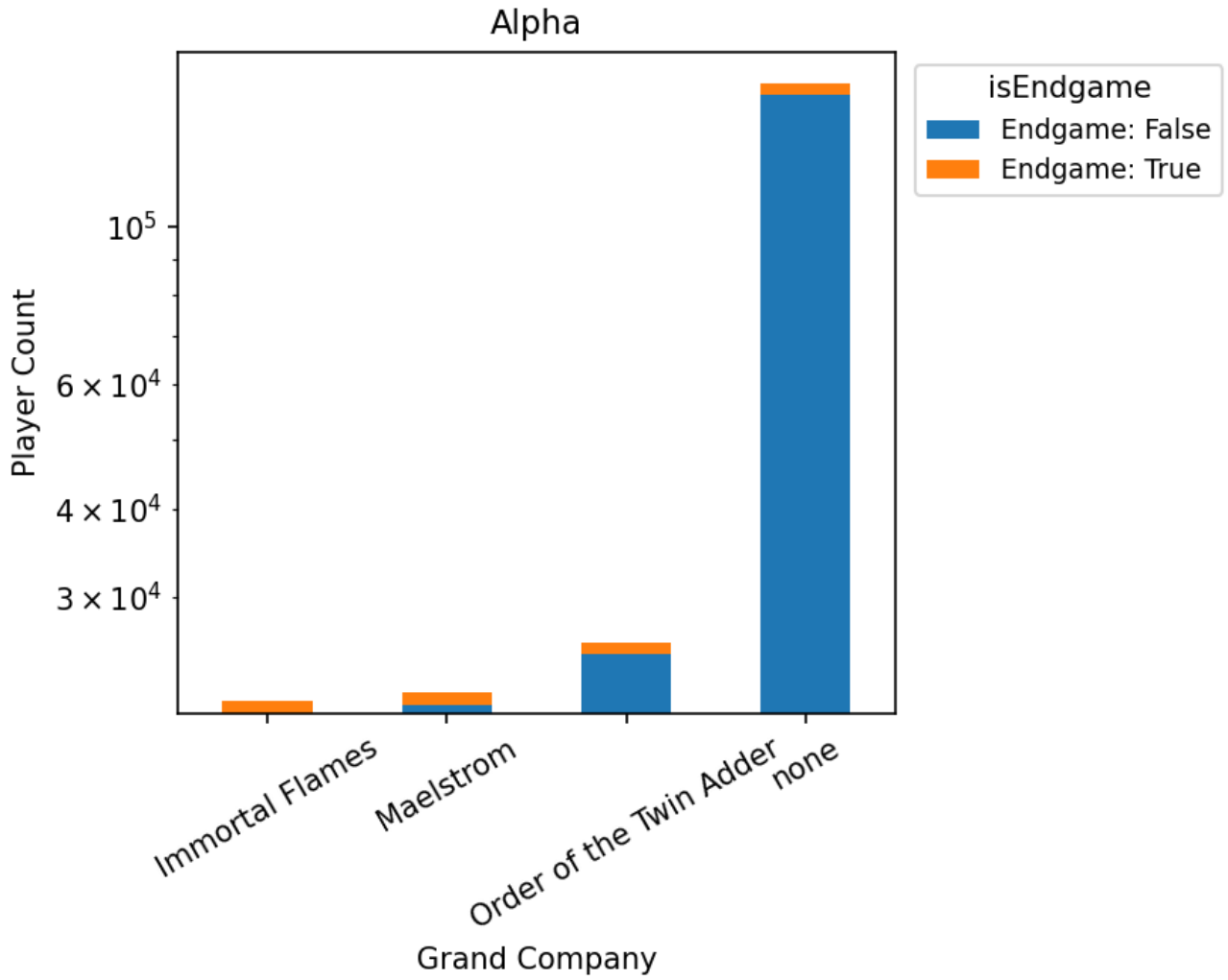
Assumption

The assumption was that all three companies should have roughly equal amounts of players.

Result

The vast majority are not part of any grand company, while the company "Order of the Twin Adder" is in second place with a significantly bigger player count than the other 2.

Plot



7.1.2. Question 2

Discussion

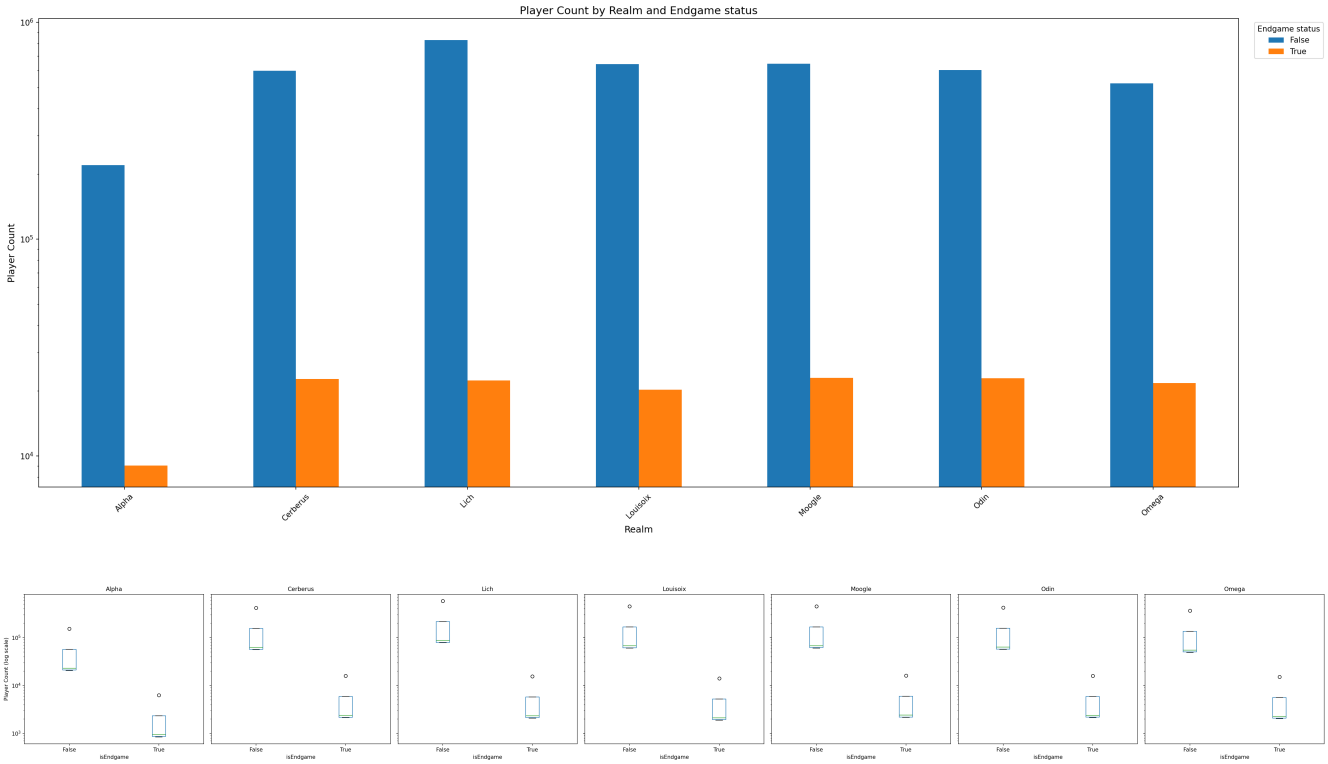
Assumption

The proportion between endgame and general player amount should be similar between all servers.

Result

It is for all servers except for the server "Alpha" which was a newly opened server at the time, so a lower portion of endgame players was expected

Plot



7.1.3. Question 3

Discussion

Result

There are 16 realms on European servers.

7.2. Ridiculous Research Questions

7.2.1. Question 1

Discussion

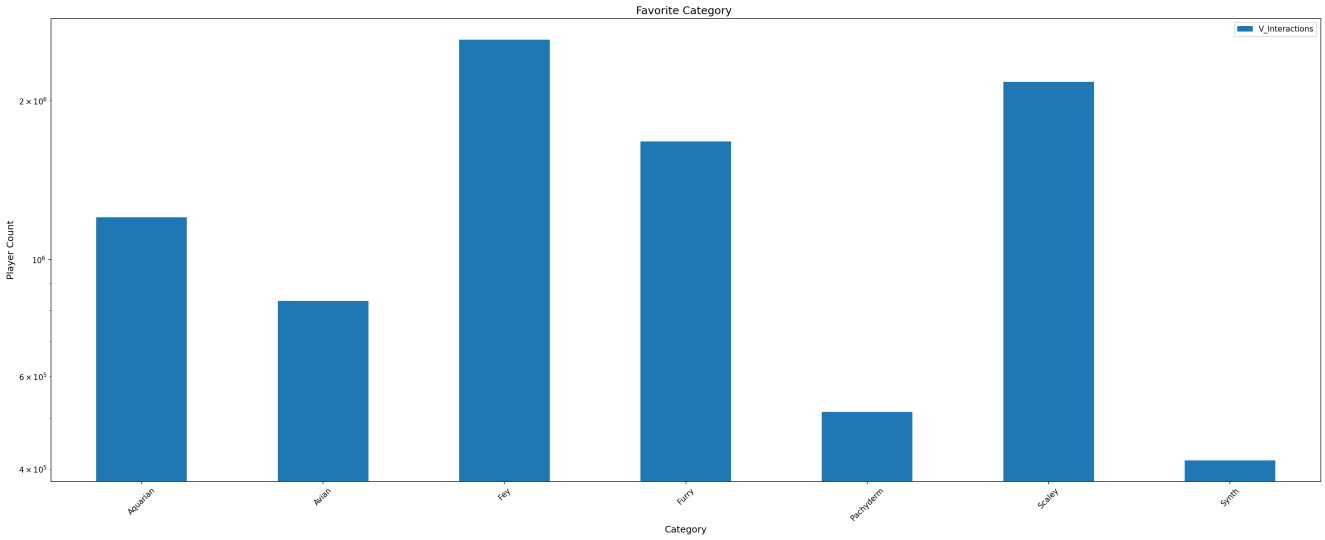
Assumption

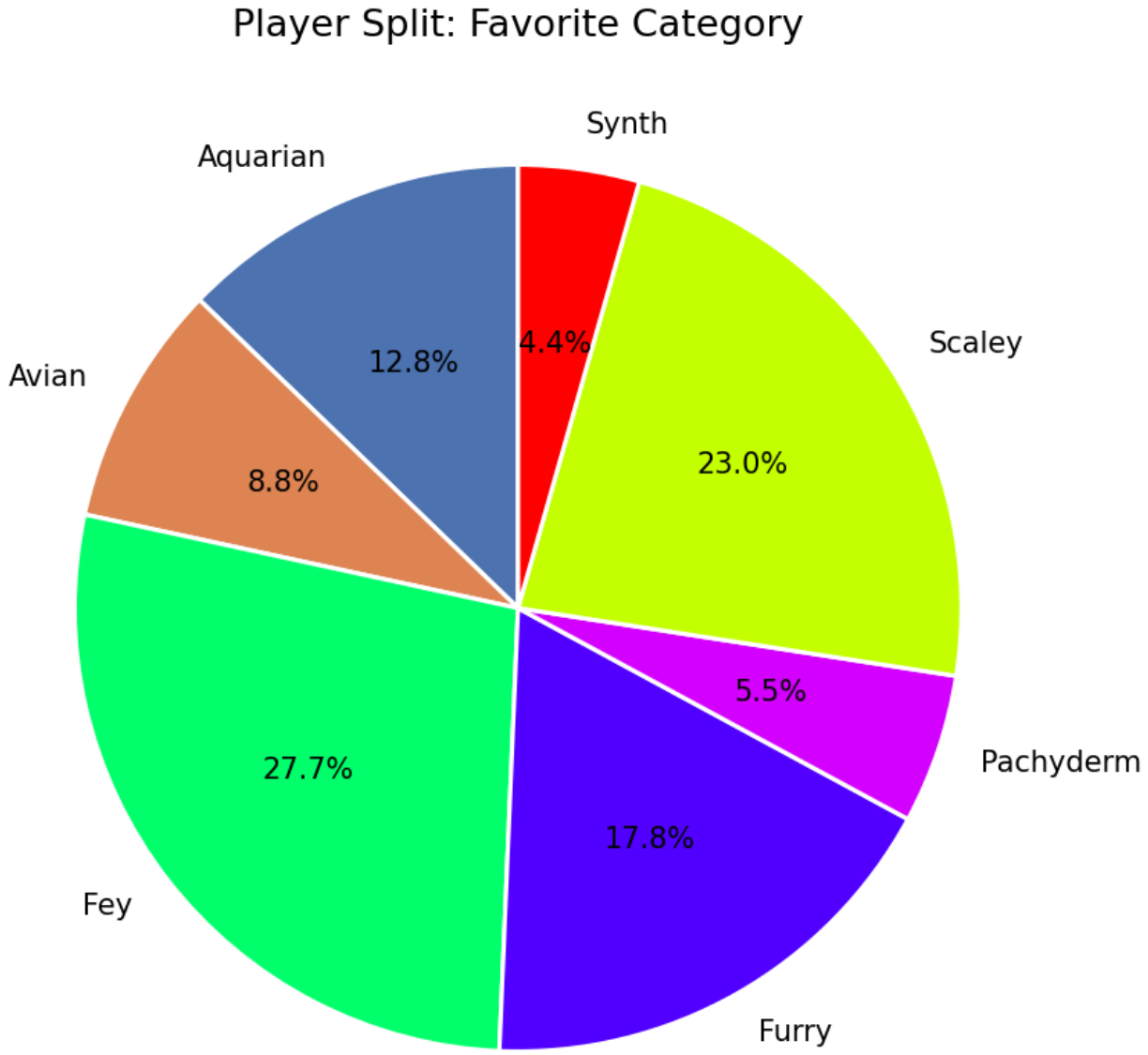
The assumption was that furry tribes are the most popular kind of furry subcategory since it is the most standard furry subcategory.

Result

The most popular category turned out to be fey's, closely followed by scalies

Plot


Favorite Category
V_Interactions
Player Count
2×10^6
10^6
6×10^5
4×10^5
Aquarian
Avian
Fey
Furry
Pachyderm
Scaley
Synth
Category


Player Split: Favorite Category
Synth
Aquarian
Scaley
Avian
Pachyderm
Fey
Furry
4.4%
12.8%
23.0%
8.8%
5.5%
27.7%
17.8%